

Julian Henry

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Work Experience

Senior Full-Stack Engineer

October 2024 – Present

Macquarie Financial Ltd. — Houston, TX · Full-time

- Deploy and operate Java servlet services powering high-priority trading workflows across globally distributed infrastructure in London, São Paulo, and Sydney.
- Translate as-dealt deal logic from C++ to Python for ASIC regulatory reporting, enabling auditable trade exposition and compliance-ready deal state.
- Co-author a shared library that adds standardized risk and workflow controls to trades across 100+ commodities and financial products.
- Deliver C++17 enhancements inside a 1M+ line enterprise trading platform, hardening low-latency, high-availability commodities and markets systems.

Software Engineer

May 2023 – October 2024

Allis-Gleaner Corporation (AGCO) — Atlanta, GA · Full-time

- Hardened AWS Kafka-based IoT telemetry infrastructure serving a connected agriculture fleet across 140 countries with improved real-time reliability.
- Engineered RESTful APIs for hardware device orchestration that accelerated integration test cycles by 500%+ across connected farm equipment.
- Shipped Angular features for the Fendt.One precision agriculture platform supporting 10,000+ remote farm operations worldwide.

Data Engineer

August 2020 – August 2022

SimSpace Corporation — Boston, MA · Full-time

- Evaluated ML-based threat detection models through adversarial red-team simulations, surfacing failure modes before production exposure.
- Built cybersecurity analytics pipelines with NLP that improved threat detection rates by 72% across enterprise security workloads.
- Led performance optimization of a 30,000+ line Python/Pandas ETL pipeline, delivering 10× throughput gains for cybersecurity data processing.

Assets & Liabilities Intern

June 2019 – May 2020

Transamerica Corporation — Baltimore, MD · Internship → part-time

- Validated actuarial risk models for variable annuity portfolios using NumPy, scikit-learn, and pandas across large-scale simulation runs.
- Achieved 99.7% simulation accuracy for reconciliation and stress testing on a \$6B+ AUM variable annuity book.

Technical Skills

Languages: C++17, Python, Java, TypeScript, JavaScript, SQL, Julia, Rust, WebAssembly

Frontend: React, Next.js, Angular, HTML5, CSS3

Backend: Spring Boot, Node.js, Express, REST APIs, Java Servlets

Cloud / Infra: AWS, Kafka, Docker, Linux, GitHub Actions, CI/CD, Cloudflare

Databases: PostgreSQL, MongoDB, DynamoDB, Neo4j, SurrealDB, SQLite, FAISS, Vector DBs

Data / ML: Pandas, NumPy, scikit-learn, TensorFlow, PySpark, RAG, Tableau

Education & Certificates

Bachelor's of Applied Science in Computer Science, University of Pennsylvania

- Vice President, Artificial Intelligence Society; undergraduate thesis on differential privacy under Rajeev Alur, with distinction
- **DELE B2** (Instituto Cervantes) — official upper-intermediate Spanish language credential

Open Source & Portfolio

- 1wg.ai — AI-powered global news gazette using RAG over a vector database for retrieval-augmented journalism across world events.
- glottosphere.com — multilingual dictionary spanning 200+ languages, powered by FAISS similarity search and a custom binary lexicon engine.
- ghee — 616+ fuzzy-searchable shell shortcuts published to PyPI; tatuagem — ASCII code-signature suite; readme-rosetta — Ollama-powered README translation.
- intersekt — WebAssembly Karatsuba vs JS BigInt, demonstrating $O(N^{1.58})$ vs schoolbook multiply.
- quantum-plankton-ml — quantum machine learning on plankton imagery across architectures.

- [laciyo](#) — constructed Romance language via simulated annealing (Python prep + Rust optimizer).